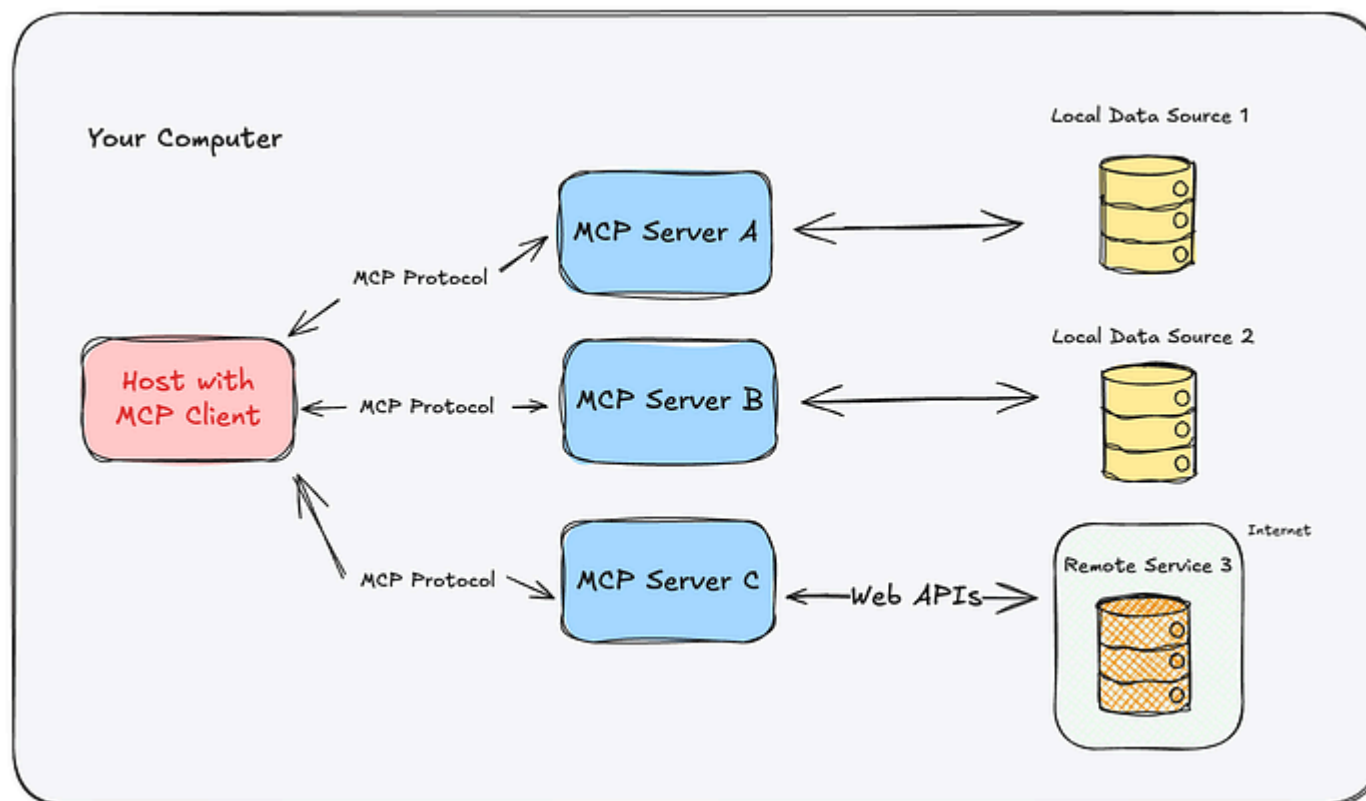


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MCP Explained: The New Standard Connecting AI to Everything

How Model Context Protocol is making AI agents actually do things



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AI agents can write code, summarize reports, even chat like humans — but when it's time to actually *do* something in the real world, they stall.

Why? Because most tools still need clunky, one-off integrations.

MCP (Model Context Protocol) changes that. It gives AI agents a simple, standardized way to plug into tools, data, and services — no hacks, no hand-coding.

With MCP, AI goes from *smart...* to *actually useful*.

What Is MCP, Really?

Model Context Protocol (MCP) is an open standard developed by Anthropic, the company behind Claude. While it may sound technical, but the core idea is

As highlighted in Forbes, MCP is a big leap forward in how AI agents operate. Instead of just answering questions, agents can now perform useful, multi-step tasks — like retrieving data, summarizing documents, or saving content to a file.

Before MCP, each of those actions required a unique API, custom logic, and developer time to glue it all together.

Traditional APIs VS Model Context Protocol

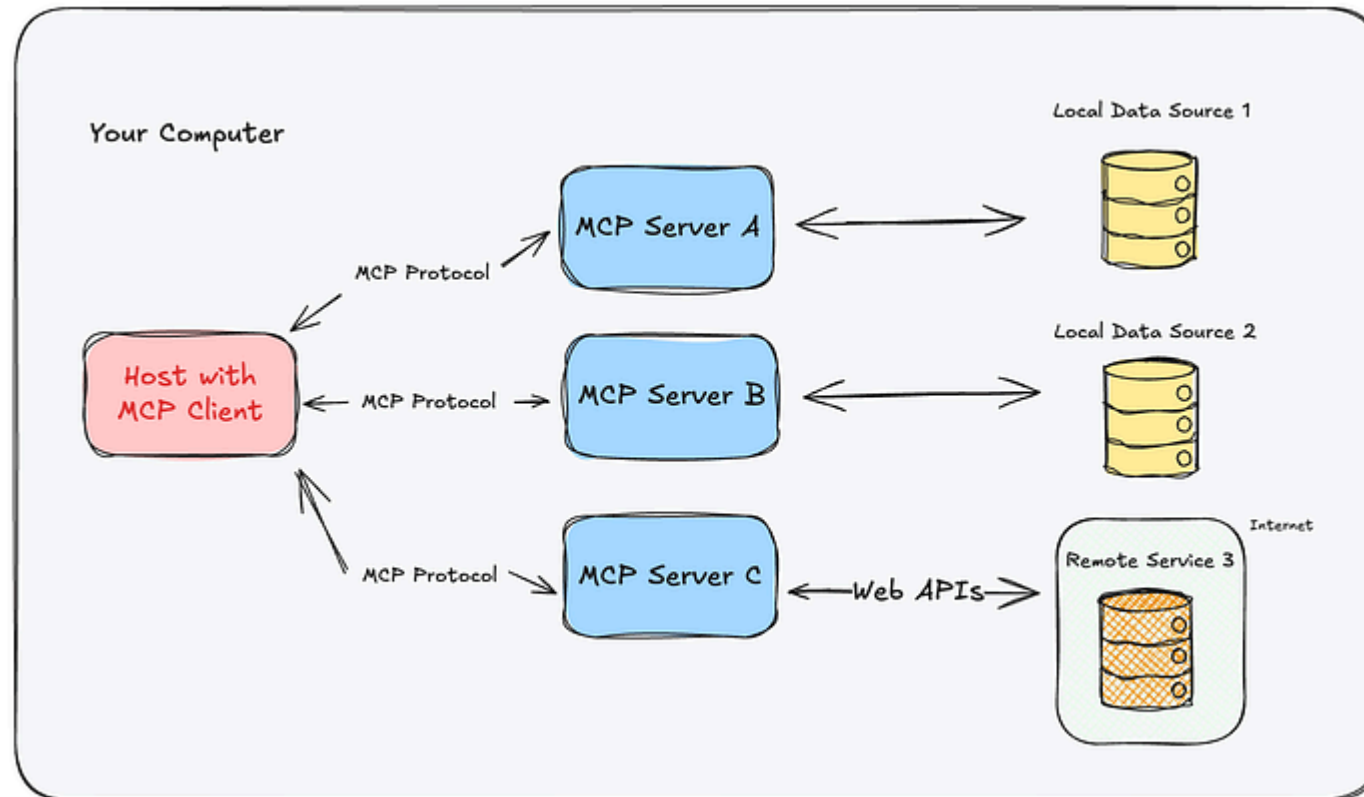
Aspect	Traditional APIs	MCP
Setup	Manual, one by one	One standard for all tools
Flexibility	Fixed, tool-specific	Dynamic and adaptable
Reuse	Hard to reuse across agents	Easy to reuse everywhere
Scalability	Breaks as systems grow	Built to scale
Agent Compatibility	Needs custom logic per tool	Works out of the box with schema
Tool Discovery	Manual configuration	Automatic, real-time

@elisowski

With MCP, it's plug-and-play. Agents can send structured requests to any MCP-compatible tool, get results back in real time, and even chain multiple tools together — without needing to know the specifics ahead of time.

In short: MCP replaces one-off hacks with a unified, real-time protocol built for autonomous agents.

The Architecture of MCP



Here is a look at how MCP works under the hood:

- MCP Host (on the left) is the AI-powered app — for example, Claude Desktop, an IDE, or another tool acting as an agent.

- Some servers access local resources (like a file system or database on your computer).
- Others can reach out to remote resources (like APIs or cloud services on the internet).

All communication between host and servers happens over the standardized MCP Protocol, which ensures compatibility and structured responses.

MCP Servers

An MCP server is like a smart adapter for a tool or app. It knows how to take a request from an AI (like "Get today's sales report") and translate it into the commands that tool understands.

For example:

- A GitHub MCP server might turn "list my open pull requests" into a GitHub API call.
- A File MCP server might take "save this summary as a text file" and write it to your desktop.

MCP servers also:

- Tell the AI what they can do (tool discovery)
- Interpret and run commands
- Format results the AI can understand
- Handle errors and give meaningful feedback

MCP Clients

On the other side, an MCP client lives inside the AI assistant or app (like Claude or Cursor). When the AI wants to use a tool, it goes through this client to talk to the matching server.

For example:

- Cursor can use a client to interact with your local development environment.
- Claude might use it to access files or read from a spreadsheet.

The client handles all the back-and-forth — sending requests, receiving results, and passing them to the AI.

The MCP protocol is what keeps everything in sync. It defines how the client and server communicate — what the messages look like, how actions are described, and how results are returned.

It's super flexible:

- Can run locally (e.g., between your AI and your computer's apps)
- Can run over the internet (e.g., between your AI and an online tool)
- Uses structured formats like JSON so everything stays clean and consistent

Thanks to this shared protocol, an AI agent can connect with a new tool — even one it's never seen before — and still understand how to use it.

Services = Real Apps and Data

The last part of the puzzle is the services — the actual tools or data sources the AI wants to use.

These could be:

Local: files on your device, a folder, an app running locally

MCP servers are the gateway to these services, handling access securely and reliably.

The MCP Ecosystem Is Taking Off

MCP is becoming a movement. What started as a developer tool is quickly turning into the backbone of how AI agents connect to the real world.

We're seeing more tools, more companies, and even entire marketplaces pop up around it. Here's what's happening.

Who's Already Using MCP?

- ❶ Block is using MCP to hook up internal tools and knowledge sources to AI agents.
- ❷ Replit integrated MCP so agents can read and write code across files, terminals, and projects.
- ❸ Apollo is using MCP to let AI pull from structured data sources.

⑤ Microsoft Copilot Studio now supports MCP too — making it easier for non-developers to connect AI to data and tools, no coding required.

Marketplaces Are Here

Here are the ones to watch:

mcpmarket.com — A plug-and-play directory of MCP servers for tools like GitHub, Figma, Notion, Databricks, and more.

mcp.so — A growing open repo of community-built MCP servers. Discover one. Fork it. Build your own.

[Cline's MCP Marketplace](#) — A GitHub-powered hub for open-source MCP connectors anyone can use.

This is the new app store — for AI agents.

Infra Tools Are Making MCP Even Easier

Mintlify, Stainless, Speakeasy → auto-generate servers with just a few clicks

Cloudflare, Smithery → make hosting and scaling production-grade servers simple

Toolbase → handles key management and routing for local-first setups

Want to Go Deeper?

Here are some great places to explore MCP further:

- [Introducing the Model Context Protocol](#) by Anthropic
- [Model Context Protocol](#) on GitHub

If you're exploring how to integrate AI agents into your workflows — [Addepto](#) can support you.

[#mcp-protocol](#) [#mcp-server](#) [#model-context-protocol](#) [#mcp-client](#) [#ai-agent](#)